

The Problem

18 SECONDS

until clinicians are forced to **interrupt** their patient

(Coyle et al., 2022)

The Problem



> 30%

of patient visits are spent
on medical history taking

(Borekci et al., 2017)



27 %

of their time is genuine
patient interaction

(Sinsky et al., 2016)



20 patients

are seen by clinicians per
day

Cascading Effect

PER WEEK



7.5^{hrs}

LOST TO INTAKE OVERHEAD



100

PATIENTS INTERRUPTED



73

VISITS: SCREEN OVER PATIENT

CUMULATIVE BURDEN



PER MONTH



30^{hrs}

= 1 FULL WORK WEEK, EVERY MONTH



400

PATIENTS INTERRUPTED



292

VISITS: SCREEN OVER PATIENT

CUMULATIVE BURDEN



But What If We Could Change This?

CLARA



Clarity before the visit.

Team



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Meet CLARA

Transforming patient conversations into FHIR-standard clinical evidence

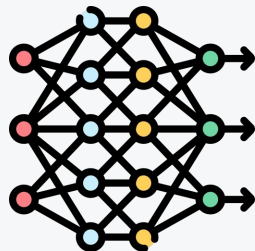
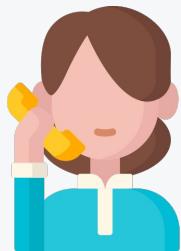
CLARA



Structured clinical interview
using flowchart-based NLP

Visual pre-visit brief to
physician's dashboard

Full attention to the patient



-24 hrs

-5 min

In room

CLARA Flow



```
Clinical Intake Summary
-----
Call SID: CAbc3b8be1c4af76371f24467859ff53c7
Patient ID: 2894
Demo Case: pneumonia_risk
Chief Complaint: Worsening cough
Symptoms: Cough with yellow mucus
Duration / Onset: Cough for one week, worsening over the last three days
Severity / Progression: Not clearly stated
Associated Symptoms: None reported (no fever, wheezing, or shortness of breath)
Relevant Past History: Chronic sinusitis, acute bronchitis
Medications Mentioned: Acetaminophen, Dextromethorphan, Doxylamine, Penicillin,
Ibuprofen, Enoxaparin
Allergies Mentioned: Not clearly stated
Red Flags: Not clearly stated
Possible Concern: Viral infection or bronchitis
Recommended (Clinician Focus: Evaluate for respiratory infection; consider
current symptoms in context of past bronchitis and sinusitis history.)

Call Summary
-----
Call SID: CAbc3b8be1c4af76371f24467859ff53c7
Patient ID: 2894
Demo Case: pneumonia_risk
Started: 2025-03-18T21:56:59.099513+00:00
Ended: 2025-03-18T20:00:21.189591+00:00

Transcript
-----
PATIENT: It's So I've had a cough for 3 days, and it is getting worse.
CLARA: Thank you for sharing that, Tish. Besides the worsening cough,
are you experiencing any other symptoms, like fever, wheezing, or shortness of breath?
PATIENT: No. Honestly, no. It's mainly just there's some mucus.
CLARA: I understand. So, you're primarily dealing with a worsening cough and some mucus
production. Has the mucus changed in color or consistency at all?
PATIENT: Some yellow mucus that comes up in a cough.
CLARA: Thank you for that detail, Tish. Just to clarify, you're noticing that the mucus is yellow,
and it sounds like it's happening when you cough.
PATIENT: No. Not really. So, like, basically, what happened was that last week, I think, I went to
Joe's Pizza in Harvard Square with 1 of my friends named Bob.
and so I've been there, and the pizza was so good. The pizza was so good.
Like, great pizza.
CLARA: I appreciate you sharing your experience at Joe's Pizza, Tish!
```

Screenshot from our working prototype



Dashboard with Epic Integration

The screenshot displays the Epic EMR interface for patient Tish Lemke (MRN 1000220020, Room 4119). The top navigation bar includes links for In Basket, Home, Change Context..., Schedule, Intra-op Board, Patient Station, My SmartPhrases, My Tools, WatchX, Online Help, Print, Log Out, Playground, and EpicCare. The patient's name and MRN are shown in the top left, along with a status indicator (green checkmark) and a red alert icon. The main content area is divided into several sections:

- Chart Review:** A tabbed interface showing various medical records. The 'Notes' tab is active, displaying a list of recent notes with columns for Date of Service, Type, and Author. The notes include Initial Assessments, H&P, ED Provider Notes, Plan of Care, Discharge Summary, and Progress Notes.
- Recent Notes:** A section showing the most recent notes, including Initial Assessments, H&P, ED Provider Notes, Plan of Care, Discharge Summary, and Progress Notes.
- 6 Months Ago:** A section showing notes from 6 months ago, including Progress Notes and Patient Instructions.
- 1 Year Ago:** A section showing notes from 1 year ago, including Progress Notes.
- Initial Assessments:** A section providing a detailed overview of the patient's initial assessment, including a brief description of the events leading to admission, patient and/or family concerns, and a list of safety alerts.
- Care Timeline:** A section showing the patient's care timeline, including admission and discharge dates.

The patient's demographic information and medical history are also visible on the left side of the dashboard.

Patient Information:

- Name:** Tish Lemke
- Gender:** Female
- Age:** 19 y.o.
- DOB:** 8/13/2004
- MRN:** 1000220020
- Language:** English
- Location:** VPMC Surg/Tn-4018-01
- Room:** 4119 Master Pt. Rm
- Code:** FULL (no ACP docs)
- Primary Ceg:** Sliding Fee/Slid...

Medical History:

- Allergies:** No Known Allergies
- DMITTED:** 6/19/2024
- Patient Class:** Inpatient
- Asthma (J45.909)**
- COVID-19 Vaccine:** Unknown
- Relations:** None
- Height:** 165.1 cm (5' 5")
- Weight:** 61.7 kg (136 lb)
- MI:** 22.63 kg/m²

Lab Results (LAST 36H):

- Lab (7)
- Imaging (1)

Active Meds (24):

- Scheduled (8)
- Continuous (4)
- PRN (6)
- Other (5)

Problems:

- Asthma (J45.909)
- Hypertension (I10)
- Type 2 Diabetes (E11.9)

Initial Assessments:

Linus Lumen, RN
Registered Nurse
Nursing

Initial Assessments
Signed

Date of Service: 6/19/2024 10:

Initial Assessments

Tish is a 19 y.o. admitted for Asthma [J45.909].
Admission type: ER.

Brief description of the events that led to the admission: shortness of breath
This information was provided by: patient

Patient and/or family denies frequent admissions due to inability to meet the expense of medication
Patient and/or family expressed the following concerns that are troubling the patient while being hospitalized:
None

Tish's primary language is English. The family speaks English. Services needed include: none. An interpreter is not required to facilitate communication with Tish, her family and other support systems.

Patient was oriented to: BRIER light, bed controls, TV/radio, patient education channel, telephone, bathroom, visiting hours, overnight stay, and call light.
The following Safety Alerts were identified during the admission process. None. Yellow "At Risk" band applied: NA
The patient and/or family reported the following note.
care for this patient. rva
Belongings with patient upon arrival/transfer; EM/wam Brrring.

Care Timeline

EO to Hosp-Admission (Current) on 6/19/2024 Pinned to Bookmarks

CLARA Impacts



**4.5 min
/ visit**

More Patients



**Targeted
Questions**

Better History



**Reduced
Rates**

Fewer Missed Diagnoses

Price Breakdown

Tier	Calls	Price
Clara Starter	300 call/mo	\$279 physician/mo
Clara Clinical	400 call/mo	\$359 physician/mo
Clara Enterprise	Unlimited	\$529 physician/mo

1 completed call (picked up and ended) = 1 patient

TAM - SAM - SOM

TAM

\$4.8B ARR

~1M US Physicians @ \$400/mo avg

SAM

~\$1.68B ARR

~350K Outpatient Physicians @ \$400/mo

SOM

~\$120M+ ARR

~25K Physicians by Year 5

Competitive Positioning

Feature	Nuance DAX	Notable Health	Clara ✦
Timing	During Visit	Pre-Visit	Pre-Visit
Red Flag Detection	✗	✗	✓
Explainable AI Model	✗	✗	✓ Cited Dx
Epic Integration	✓	✓	✓ FHIR

ROI Calculation

Metric	Primary Care	Internal Medicine	Pulmonology	GI (Medical)
Method 1 — Time Value (Salary-Based, Conservative)				
Value of time saved per successful call (\$)	\$9.38	\$11.25	\$13.50	\$16.88
— Start tier (\$229): calls needed to break even	24.4	20.4	17.0	13.6
— Start tier: break-even PICKUP RATE needed	7.0%	7.3%	6.1%	5.4%
— Clinical tier (\$399): calls needed to break even	42.6	35.5	29.6	23.6
— Clinical tier: break-even PICKUP RATE needed	12.2%	12.7%	10.6%	9.5%
— Enterprise tier (\$549): calls needed to break even	58.6	48.8	40.7	32.5
— Enterprise tier: break-even PICKUP RATE needed	16.7%	17.4%	14.5%	13.0%

Timeline

**Phase 1:
MVP
Development**



**Phase 3:
Expansion to
Primary Care**



**Phase 2:
Pilot testing in
Pulmo Clinics**



**Phase 4:
More Features
for Specialities**



8

physicians ready to use

Clinical Impact



**100 minutes
saved daily**



**Reduced
no-shows**



**Increased EOC scores &
patient satisfaction**

**Help us untangle patient
stories**

CLARA



Appendix

Appendix TOC

- Clara Intelligence Layer
- EPIC/FHIR Integration
- Technical Workflow
- HIPAA Compliance
- FDA
- Incentives
- Cost Breakdown
- Scaling
- Competitive Positioning
- CPT Codes
- ROI Calculations

Clara Intelligence Layer

A Conversation Engine

Real-Time Voice AI

PROMPT ARCHITECTURE

- Clinical interviewer system prompt
- Dynamic context injection: **FHIR preload**, visit type, demographics

FLOW CONTROL (STATE MACHINE)

CC → HPI → ROS → Meds → Allergies

- Adaptive **branching logic** per response

TURN HANDLING

- Barge-in interruption supported
- **~1.5s** silence → turn boundary
- Clarification loops for ambiguity

OUTPUT

Speaker-separated transcript + Timestamps

B Clinical Extraction Layer

LLM + Schema Validation

Raw Transcript → Structured JSON

SCHEMA ENFORCEMENT

- Pydantic validation · missing fields flagged
- Extracts: onset, duration, severity, **negatives**

```
{
  "chief_complaint": "Chest pain",
  "duration": "2 days",
  "severity": "7/10",
  "assoc_symptoms": ["SOB", "fatigue"],
  "negatives": ["no fever", "no cough"],
  "medications": ["lisinopril"],
  "red_flags": []
}
```

C Red Flag Detection

Hybrid Safety Layer

RULE-BASED TRIGGERS

- **Chest pain + SOB** → cardiac risk
- **Unilateral weakness** → stroke alert

LLM CONTEXTUAL REASONING

- Detects subtle, vague patterns
- Context-aware — not keyword matching

PRIORITY SCORING

Low Moderate High

LIVE EXAMPLE

● Chest pain + SOB → HIGH RISK · Cardiac

OUTPUT

- Highlighted alerts pushed to **physician dashboard**

D Case Matching

Clinical Context Retrieval

EMBEDDING MODEL

- **text-embedding-3-small** (OpenAI)
- Alt: BioMedBERT for clinical domain

VECTOR SEARCH

- Top-K similar cases · cosine similarity

RETRIEVAL PIPELINE

Case → Embed → Vector DB → Top-K

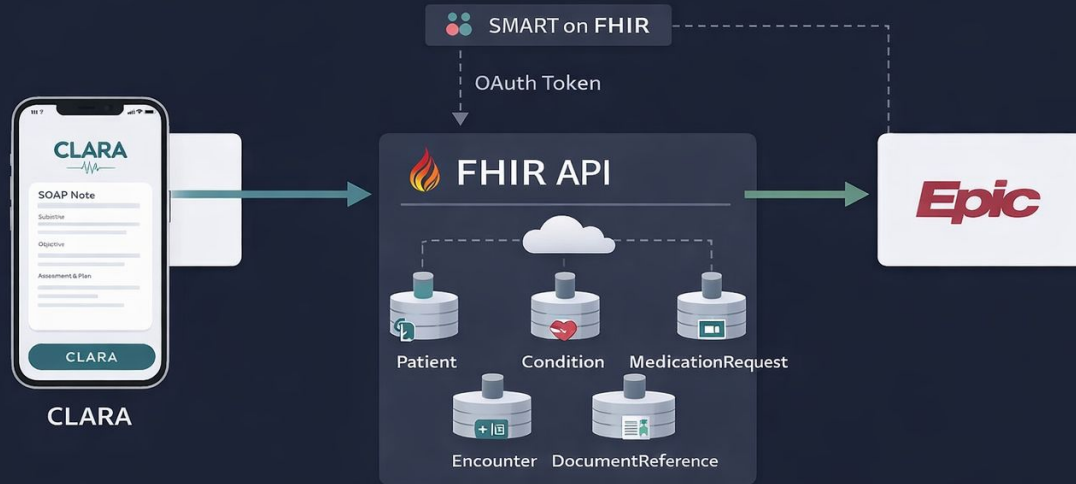
KNOWLEDGE SOURCES

- Synthetic cases + curated clinical datasets
- Future: real de-identified system data

! "Pattern recognition, not diagnosis"

Epic/FHIR Integration

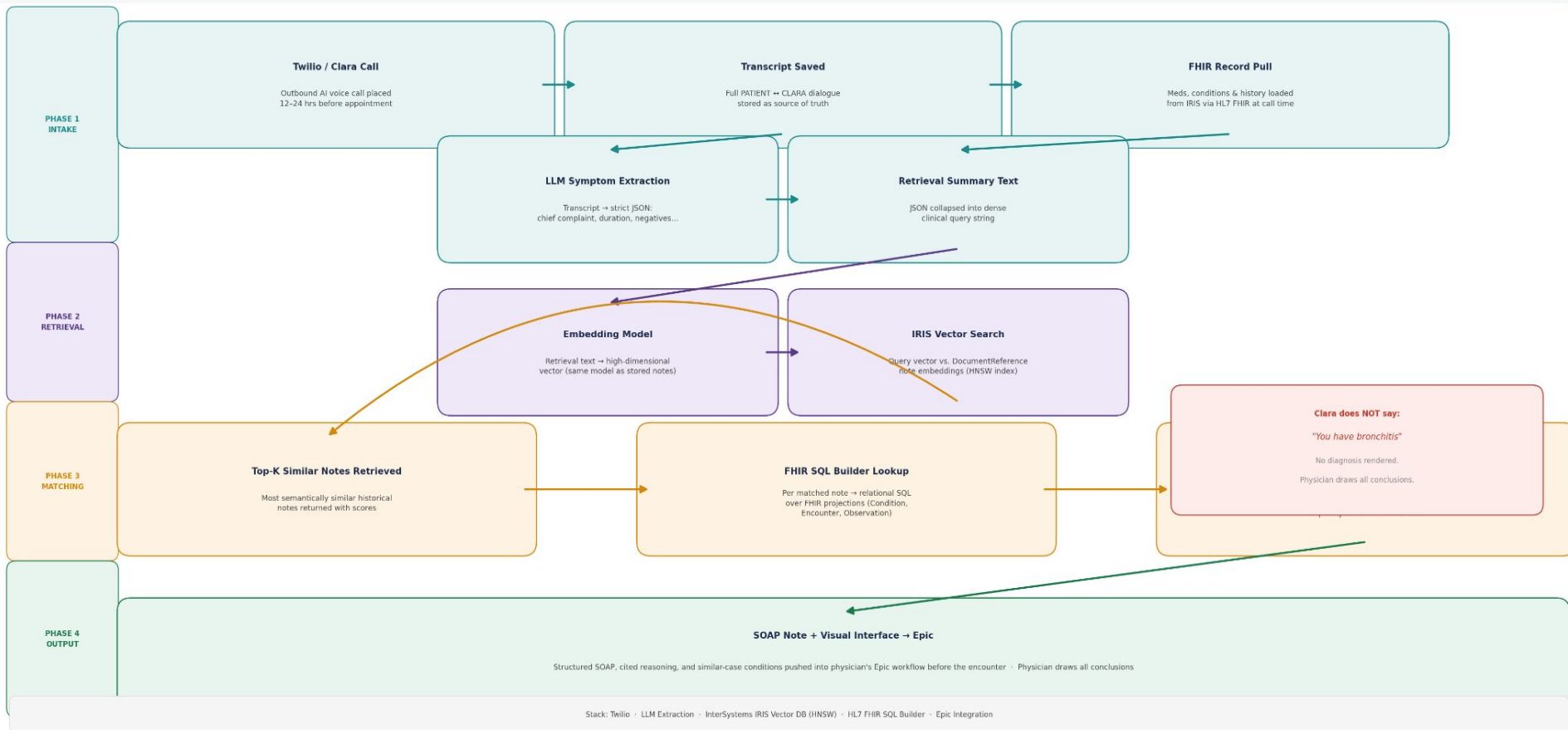
CLARA's Epic / FHIR Integration



FHIR Resources Used:

- | | | | |
|--|---|---|--|
|  Patient |  Condition |  MedicationRequest |  Encounter |
|  Condition | | |  DocumentReference |

Technical Workflow



HIPAA Compliance

01

Business Associate Agreement (BAA)

A signed BAA must be executed with every covered entity (hospital, clinic, practice) before any PHI is accessed or processed.

Required

02

Encryption in Transit & at Rest

All PHI transmitted via FHIR must use TLS 1.2+ (HTTPS). Stored data must be encrypted at rest (AES-256 or equivalent).

Required

03

Access Controls & Minimum Necessary

SMART on FHIR / OAuth 2.0 limits Clara to only the records required for intake. No broad chart access permitted.

Required

04

Audit Logging & Breach Notification

Every PHI access must be logged with timestamp and user. Breaches require patient notification within 60 days.

Required

FDA

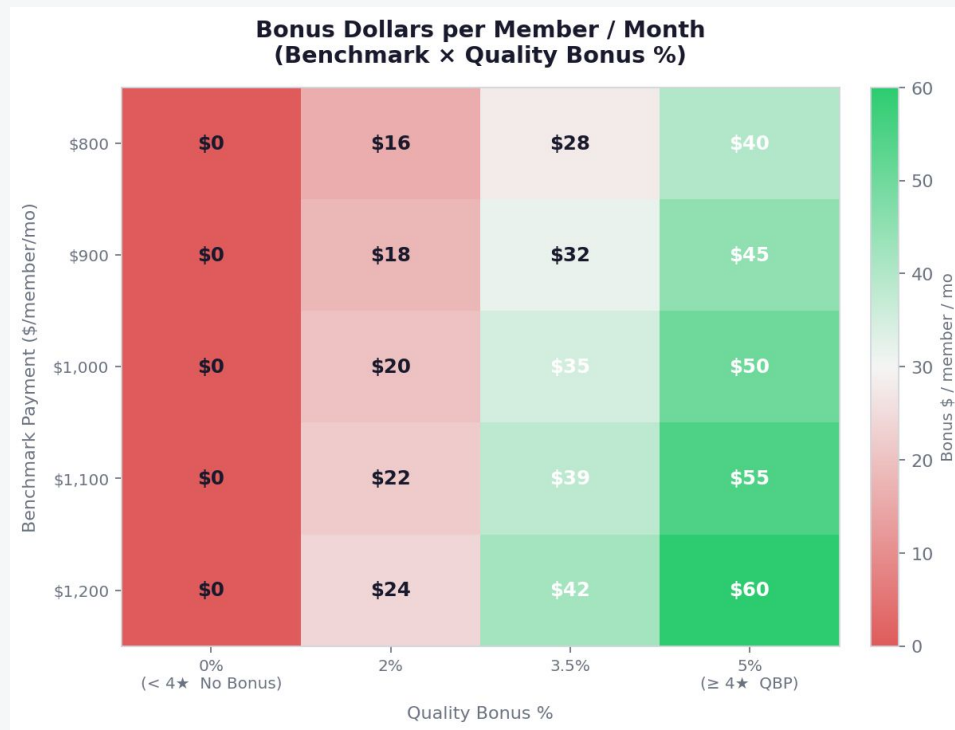
Source: FDA Final Guidance — Clinical Decision Support Software (Sept. 28, 2022) · 21st Century Cures Act, FD&C Act §520(o)(1)(E)

✓ **Clara meets ALL FOUR criteria for Non-Device CDS and is EXEMPT from FDA medical device regulation**

FDA CRITERION (§520(o)(1)(E))	HOW CLARA SATISFIES THIS CRITERION
Criterion 01 No Medical Imaging or Signal Processing Not intended to acquire, process, or analyze a medical image, in vitro diagnostic signal, or physiological pattern/signal.	✓ Clara Qualifies ▶ Clara processes spoken language and structured text from patient calls. It does not acquire, process, or analyze any medical imaging, lab signals, or biosensor data of any kind.
Criterion 02 Displays Medical Information Intended to display, analyze, or print medical information about a patient, or other medical information such as clinical studies and practice guidelines.	✓ Clara Qualifies ▶ Clara surfaces structured patient history, symptom clusters, and matched historical case patterns to the physician inside Epic — displaying medical information to support the clinical encounter.
Criterion 03 Supports HCP Recommendations Intended to support or provide recommendations to a health care professional about prevention, diagnosis, or treatment of a disease or condition.	✓ Clara Qualifies ▶ Clara presents similar historical cases to the physician — it does not render a diagnosis. It informs the physician's judgment; it does not replace it. The clinician draws all conclusions.
Criterion 04 Clinician Can Independently Review Basis Intended to enable the health care professional to independently review the basis for the recommendations so they are not required to accept it.	✓ Clara Qualifies ▶ Every case match in Clara shows exactly which symptoms and history elements drove the match. Physicians see full reasoning — they can accept, modify, or ignore any suggestion at any time.

Incentives

- Reimbursements
- Patient Satisfaction
- Clinical Compliance
- Efficiency



Cost Break Down

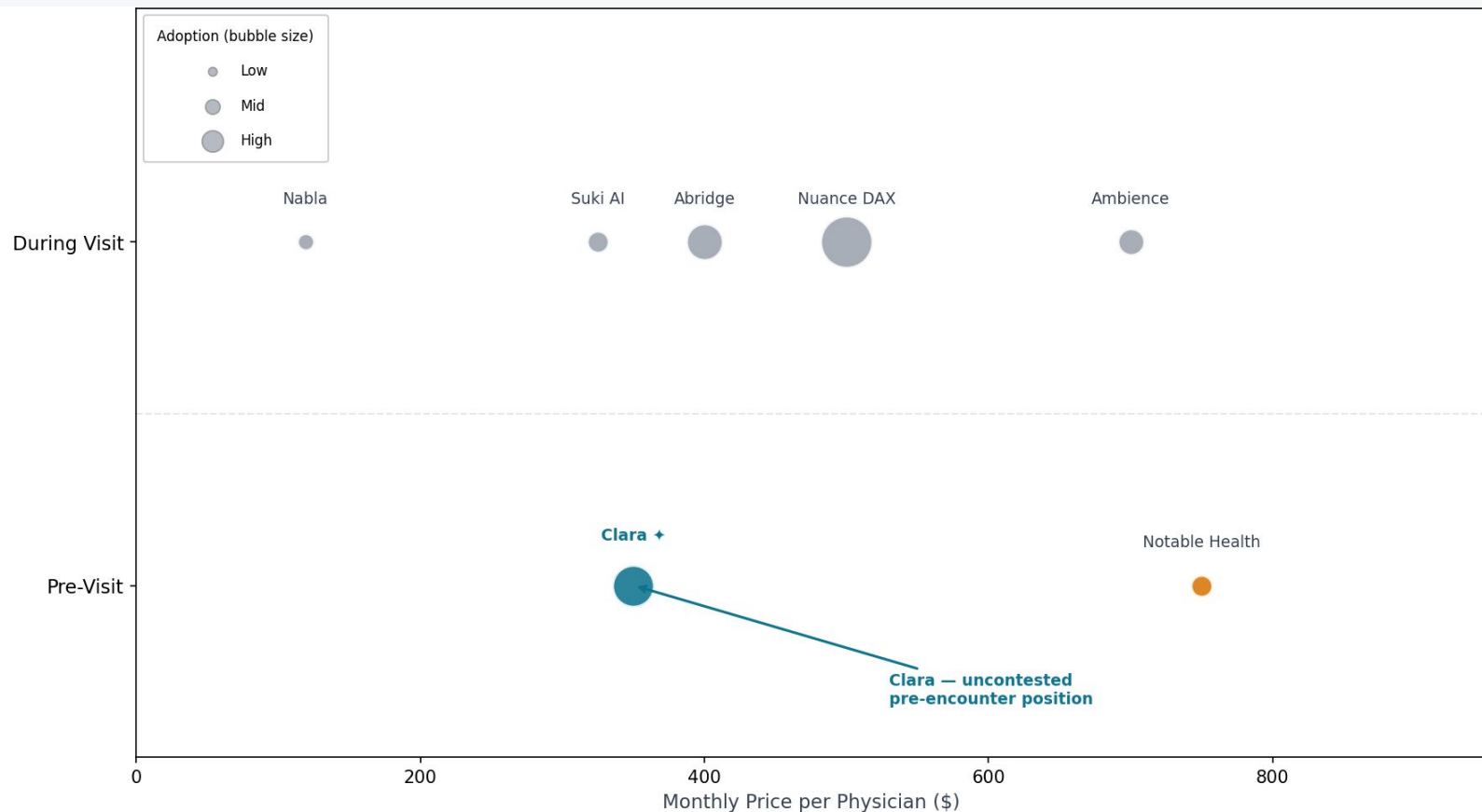
PER-CALL COST BREAKDOWN				
Cost Component	Avg Call (Early Stage)	Complex Call (Early Stage)	Avg Call (At Scale)	% of Avg Cost (Early)
Twilio Outbound Voice	\$0.130	\$0.234	\$0.090	48.0%
Speech-to-Text Transcription	\$0.060	\$0.108	\$0.040	22.1%
LLM Inference (blended)	\$0.046	\$0.078	\$0.025	17.0%
FHIR / Epic API Queries	\$0.010	\$0.010	\$0.008	3.7%
Vector DB Search (IRIS)	\$0.005	\$0.005	\$0.003	1.8%
Infrastructure Overhead	\$0.020	\$0.020	\$0.010	7.4%
TOTAL COST PER CALL	\$0.271	\$0.455	\$0.176	100.0%

Scaling



✦ Pilot studies conducted at each phase to validate outcomes and build institutional confidence

Competitive Position



CPT Codes

● E/M OPTIMIZATION

99213	Office Visit — Low Complexity	\$78.68	Baseline; prevents downcoding under Clara documentation
99214	Office Visit — Moderate Complexity	\$110.53	+\$31.85/visit — most common upcode target via MDM support
99215	Office Visit — High Complexity	\$148.90	Red-flag detection directly supports high-complexity MDM

● CHRONIC CARE MANAGEMENT

99490	CCM — Basic (≥ 20 min/mo)	\$62.22	Clara call + FHIR pull satisfies monthly time threshold
99487	Complex CCM (≥ 60 min, 2+ Dx)	\$131.20	Multi-condition FHIR pull captures complexity automatically

● PRINCIPAL CARE MANAGEMENT

99424	PCM — Single Condition (≥ 30 min/mo)	\$75.10	Widely underbilled; Clara intake satisfies time threshold
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● TRANSITIONAL CARE MANAGEMENT

99495	TCM — Moderate Complexity (14-day)	\$167.65	Post-discharge call satisfies required patient contact
99496	TCM — High Complexity (7-day)	\$235.93	High-acuity discharge; Clara flags readmission risk

Clinical Impact



**100 minutes
saved daily**



**Reduced
no-shows**



**Decreased
clinician burden**



**Increased EOC
scores & patient
satisfaction**